CPS / CS637

Ready-to-read presenter scripts

Stealthy Computational Delay Attacks on Control Systems

Swarali Nilesh Savale

Deck slides 8-17: system model, attack optimisation and timing-based detection.

Srinjoy Chakraborty

Deck slides 34-40: policy replay and defender study. Slides 41-42: the shared group closing.

Read the numbered speech blocks aloud. Follow the green stage directions silently. Every slide has an explicit entry, pointing sequence and exit.

Includes equation walkthroughs, real-life analogies, numerical readings, hand-offs and 30 ready Q&A answers. The scope distinguishes the paper's findings, completed project work and planned experiments.

Based on the saved 42-slide LaTeX deck, checked against the paper and the supplied code/results. The presentation source is unchanged.

How to use this script

Read only the speech

The green numbered line is an action cue: where to point, when to pause, or when to click. The black paragraph immediately below it is exactly what to say. Do not announce the cue, step number, reference note or at-a-glance equation.

Follow the current deck numbers

These are slide numbers in the revised 42-frame deck, including the speaker dividers and the two glossary slides. Use each slide title as the second check. Turning a PDF page is not a cue to advance the deck: click only when the green instruction says to. The old annotated 33-slide PDF has different numbers.

Keep gestures small

Use the pointer to identify an equation, axis, curve or table row. Move it away when addressing the audience. The script specifies the important location; there is no need to trace every symbol continuously.

Let pauses do the work

The short pauses are deliberate. They give the audience time to read a matrix, compare two values or absorb a caveat. Keep the notation visible while explaining it.

Keep the Q&A separate

The answer bank is not part of the main spoken script. If asked a matching question, read the answer in that bank. Otherwise continue the numbered presentation without interruption.

Use the scripted hand-offs

Swarali ends by advancing to Madhavi’s divider, slide 18. Srinjoy starts after Abhishek, covers slides 34-40, and then delivers the shared Conclusion and References on slides 41-42.

Detailed delivery time

Swarali Nilesh Savale: approximately 26-30 minutes for the full spoken script, including brief transitions and pauses. The script contains about 3,054 spoken words.

Srinjoy Chakraborty: approximately 23-26 minutes for the full spoken script, including brief transitions and pauses. The script contains about 2,630 spoken words.

Timing estimates use 110-125 spoken words per minute. Questions and unplanned discussion add time.

Pronunciation and reading key

On the slide	Say aloud
ψ_k	“psi at k” (psi sounds like “sigh”)
Φ_h / Φ_m	“phi sub h” / “phi sub m” (phi sounds like “fie”)
y-tilde_k	“y tilde at k”
a-bar	“a bar”
p-hat-m	“p hat sub m”
ℓ	“ell”, the letter L; it is the horizon
ω / ω*	“omega” / “omega star”
χ²	“chi square” (chi sounds like “kai”)
α / φ	“alpha” / “phi”
x^T Q x	“x transpose Q x”
arg min	“the choice that minimises”
MIQCP	“mixed-integer quadratically constrained programme”
FA	“F A”, or “session false-alarm probability”
HOLD / ZERO	“hold” / “zero”; these are actuator fallback policies

Do not read underscores, multiplication signs or code spellings literally unless the spoken paragraph tells you to. The speech already translates the important equations into words.

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Swarali Nilesh Savale

Problem formulation and timing-based detection

Deck slides 8-17

Start after Shubham’s Key takeaways slide. Keep your main message in mind: the physical process keeps moving while computation can stall. Your equations make that separation explicit.

Your sequence: divider; two glossary slides; model scenario; nominal plant/controller; augmented switched model; explicit matrices; attacker objective; watchdog/binomial; geometric chi-square test.

Full delivery: about 26-30 minutes. The following pages are the complete spoken script.

Slide 8 | Problem Formulation

Swarali Nilesh Savale / Paper Sections 3-4; opening hand-off

01 [Face the audience. Keep the title slide displayed.]

Thank you, Shubham. I am Swarali Nilesh Savale. Shubham explained the attack idea. I will now show how the paper turns that idea into a mathematical problem.

02 [Point briefly to “System model, attack optimisation and timing-based detection”.]

My section answers three questions. What continues changing when the controller is late? How does the attacker choose the damaging deadlines? And what must the attack satisfy to avoid the timing tests?

03 [Keep your pointer still. Use a conversational tone.]

Imagine balancing a stick on your hand. Your last correction may have been perfectly sensible when you calculated it. But if it arrives late, the stick has already moved. The command value and the command timing both matter.

04 [Pause for two seconds.]

That is only an analogy. The paper gives us a precise linear model, a particular rule for delayed jobs, and explicit detector constraints. The conclusions apply within those assumptions.

05 [Click once to slide 9.]

Let me first introduce the symbols we will actually use, so the equations are easier to follow.

Slide 9 | Symbol glossary: plant, controller and timing

Swarali Nilesh Savale / Deck notation; paper Eqs. (1)-(7)

AT A GLANCE / do not read this block separately

```
x = physical state; y = measured output; u = applied input; z = controller memory
ψ = [x; y-tilde; z; u]; a = 0 for a hit, 1 for a miss
```

01 [Point down the left column: x, y, then u.]

Think of x as what the physical process is doing, y as what we measure, and u as what we apply. For a tank, x can contain water levels, y the measured levels, and u the pump-input deviation.

02 [Point to z, then to y-tilde.]

The controller can also have memory, which we call z. Y tilde is the measurement stored for the computation. This distinction matters because a delayed computation does not automatically receive a fresh measurement.

03 [Point to k, h and T_s.]

K labels the sampling periods. H or T sub s gives the physical duration of one period. A delay of five samples means very different physical time for a five-millisecond pendulum and a slower tank.

04 [Point to u_k and u_(k+1), in that order.]

U at k is the input already acting on the plant. U at k plus one is the next command being calculated. The paper deliberately includes this one-period command convention.

05 [Move to the right column: A,B,C,D; then F,G,H,K.]

The first four matrices describe the plant. The next four describe the controller. These capital letters are fixed matrices here; they are not the attack decisions.

06 [Point to ψ and then Φ_h, Φ_m.]

Psi collects the physical state and the computational memory into one larger state. Phi sub h moves that state forward after a deadline hit. Phi sub m moves it forward after a miss.

07 [Point to Δ, a_k and ℓ.]

Delta specifies what happens to the actuator command on a miss: HOLD or ZERO. A at k is the binary choice of hit or miss. Ell is the number of periods in the attack horizon.

08 [Point to θ; do not dwell on this row.]

Theta is the pendulum angle. Later, the plot will display degrees, while the terminal angle cost will use radians squared. We should keep those units separate.

09 [Click once to slide 10.]

The next glossary groups the detector quantities and the cost quantities. We will meet them again in the actual equations.

Slide 10 | Symbol glossary: detectors, costs and project links

Swarali Nilesh Savale / Paper Section 4.4 and Definition 5.1; project defender notation

AT A GLANCE / do not read this block separately

```
ω = J_attack - J_nominal
N = sum of the counted hit-to-hit intervals; T = N × S_chi-square
```

01 [Point to a-bar and n_w in the left column.]

A bar limits consecutive misses. N sub w is the number of jobs in the binomial window. These are different restrictions: a burst limit is not a total-count limit.

02 [Point to p-hat-m and α.]

P hat sub m is the estimated miss probability during normal operation. Alpha is the significance level for a statistical test. Neither symbol is the probability that an attack is present.

03 [Point to the binomial critical count.]

The binomial quantile gives an allowed count of misses. It converts the probability model into a threshold the monitor can actually check.

04 [Point to m_r, p_r, N and E_r, in sequence.]

The geometric test groups intervals by the number of misses between two hits. M sub r counts an observed category. P sub r is its model probability. N is the number of counted intervals, and E sub r is the expected count.

Slide 10 | continued

Symbol glossary: detectors, costs and project links. Stay on this deck slide until the explicit click cue.

05 [Point to S_chi-square and then T.]

S is the usual chi-square statistic. The code also uses T, which is N times S. Multiplying both the statistic and its limit by N changes the algebra, not the decision.

06 [Move to the right column, Q then the two J terms.]

Q chooses which output directions matter in the terminal cost. J attack is the cost with the chosen miss sequence. J nominal is the corresponding cost without the attack, from the same initial condition.

07 [Point to ω and ω-star.]

Omega is the difference between those costs. Omega star means the largest feasible difference for a fixed configuration and horizon. A particular replay and an optimised worst case are different objects.

08 [Point to d, FA and φ.]

The final symbols belong to our defender study. D is a configuration choice. F A is the probability of an alarm during a whole nominal session, and phi is its permitted limit.

09 [Point briefly to the project-links box. Do not click either link.]

The project links shown here are editable placeholders, rather than the sources of the numerical claims. The actual paper and official repeatability code are listed in the references.

10 [Click once to slide 11.]

With the notation in place, let us look at the physical situation the model is trying to capture.

Slide 11 | Part 2: the model

Swarali Nilesh Savale / Paper Sections 3.1-3.2 and 4.1

01 [Point to the controller block on the right.]

This block is a periodic controller running as a software task. Each job has a deadline because the physical process needs the next command at a predictable time.

02 [Trace the upper arrow from controller to plant.]

The controller calculates a command for the next period. Now trace the lower arrow back: the sensor measurement closes the feedback loop.

03 [Point to the attacker block and the dashed arrow.]

The attacker acts on the execution of the controller. It needs a way to pre-empt or block the job, for example sufficient scheduling priority or access to a shared resource. An arbitrary harmless background task is not automatically enough.

04 [Point to the plant block. Pause briefly.]

The attacker does not have to change the plant equations or forge the measurement. The plant keeps evolving while the controller has failed to deliver a fresh command.

05 [Look at the audience.]

Imagine a thermostat calculation being held up while the room temperature continues changing. Freezing the calculation does not freeze the room. That separation between software progress and physical progress is the key idea.

06 [Point to the modelling-scope box.]

The paper uses a linear plant, a linear controller and periodic execution. Its nominal timing model treats deadline misses as independent events with an estimated probability. Those are modelling assumptions, not universal properties of every CPS.

07 [Point to the research-question box.]

The attacker therefore has a constrained objective: cause the largest physical deviation while making the timing record acceptable to the active detector.

08 [Click once to slide 12.]

We can now describe the normal loop with two compact sets of equations.

Slide 12 | Discrete-time plant and controller

Swarali Nilesh Savale / Paper Eq. (1), Eq. (2); Section 3.1

AT A GLANCE / do not read this block separately

```
x_(k+1) = A x_k + B u_k        y_k = C x_k + D u_k
z_(k+1) = F z_k + G y_k        u_(k+1) = H z_k + K y_k
```

01 [Point to the first line of the upper-left box.]

Equation one says: the next physical state equals the state's own evolution, A x, plus the effect of the input, B u. Even when no new software calculation finishes, this physical update still happens.

02 [Point to the second line of the same box.]

The output equation says what the sensor reports. C maps the internal state to the measured output. D allows a direct input contribution to that output; in many models D is zero, but we keep the general expression.

03 [Point to the controller equation for z.]

In equation two, the controller updates its internal memory using its old memory and the measurement. This can represent a dynamic controller, such as one with an estimator or integral action.

04 [Point to the controller equation for u_(k+1).]

The next command combines controller memory and measured output. Read the subscript carefully: this is u at k plus one, not u at k.

Slide 12 | continued

Discrete-time plant and controller. Stay on this deck slide until the explicit click cue.

05 [Trace the loop on the right from plant, through feedback, through controller, and back.]

At the start of period k, the plant already has u at k. The controller samples y at k and spends this period calculating the command that will act in the next period. That is the paper's timing convention.

06 [Look at the audience; hold your hand level to suggest a constant input.]

Imagine a pump holding its present setting while a controller calculates the next setting. The physical level changes under the setting already applied, not under a command that has not yet finished computing.

07 [Point back to x and z.]

Notice also that x and z are different states. X belongs to the physical process. Z belongs to the controller algorithm. On a miss, they do not have to behave in the same way.

08 [Point to the timing-convention box.]

The goal is regulation toward zero. For a tank operating around a setpoint, zero can mean zero deviation from that operating point, rather than an empty tank.

09 [Click once to slide 13.]

The normal equations are straightforward. The interesting question is how to describe a job that does not finish on time.

Slide 13 | Computational delay attack and switched dynamics

Swarali Nilesh Savale / Paper Eqs. (3)-(6); binary attack notation in Eq. (7)

AT A GLANCE / do not read this block separately

$$\psi_k = [x_k;\ \tilde{y}_k;\ z_k;\ u_k]$$
$$\psi_{k+1} = [(1-a_k)\,\Phi_h + a_k\,\Phi_m]\,\psi_k$$

01 [Point down the four entries of ψ in the upper-left box.]

To describe a delayed job, the physical state alone is insufficient. We also need its stored measurement, controller memory and current actuator command. Together, these form the augmented state psi.

02 [Point to y-tilde.]

The stored measurement is important because this paper continues a delayed computation. It does not discard the job and restart from a fresh sensor sample after every miss.

03 [Point to the binary-decision definition.]

A at k equals zero for an on-time completion and one for a miss. These are the attacker's decisions. The attacker does not directly choose the next plant output.

04 [Point to (1-a_k) in the switched equation.]

If a at k is zero, one minus a is one, so only Phi h remains. The model takes the deadline-hit update.

05 [Point to a_k Φ_m.]

If a at k is one, the first term disappears and Phi m remains. This is a switch between two updates, not an average of half a hit and half a miss.

Slide 13 | continued

Computational delay attack and switched dynamics. Stay on this deck slide until the explicit click cue.

06 [Point to the Miss bullet on the right.]

On a miss, the physical state continues evolving. Controller memory and the stored measurement remain fixed. The actuator either holds its old command or receives zero, according to the chosen policy.

07 [Point to the Hit bullet.]

When the delayed computation finishes, it uses the stored measurement for its controller update. In the state convention on the next slide, the measurement buffer then refreshes for the next computation.

08 [Point across the five bits, starting at k=0.]

Read the example one one zero one zero as: miss, miss, hit, miss, hit. The maximum consecutive run is two, and the total number of misses is three. Those two numbers will matter to different detectors.

09 [Pause, then point back to the switched equation.]

For a fixed initial state, the chosen sequence selects a product of hit and miss matrices. The order matters: moving the same number of misses to different periods can change the final state.

10 [Click once to slide 14.]

Let us now open up those matrices. Their rows show exactly what freezes and what continues moving.

Slide 14 | Hit and miss matrices in the paper

Swarali Nilesh Savale / Paper Eq. (4), Eq. (5), Eq. (6); reconstructed expansion of the buffer row

AT A GLANCE / do not read this block separately

```
Φ_h rows: [A,0,0,B]; [CA,DK,DH,CB]; [0,G,F,0]; [0,K,H,0]
Φ_m rows: [A,0,0,B]; [0,I,0,0]; [0,0,I,0]; [0,0,0,Δ]
```

01 [Point to the state ordering at the top; then the first row of both matrices.]

The four block rows correspond to x, y tilde, z and u, in that order. Both matrices have the same first row, A, zero, zero, B. Therefore the plant evolves under its currently applied input whether the job finishes or not.

02 [Point to the bottom row of Φ_h on the left.]

Start at the bottom of the hit matrix. Multiplying this row by psi gives u at k plus one equals K times the stored measurement plus H times controller memory. That is the command produced when the computation completes.

03 [Point one row up in Φ_h.]

The third row gives z at k plus one equals G times the stored measurement plus F times the old controller state. The output command and controller memory both use the measurement associated with that computation.

04 [Point to the second row, CA, DK, DH, CB. Speak slowly.]

The second row refreshes the stored output. It looks more complicated, but it is simply the plant's output equation applied at the next step. I will expand it once.

05 [Point first to CA and CB.]

The next physical state is A x plus B u. Multiplying it by C gives C A x plus C B u.

06 [Point to DH and DK.]

The next output can also contain D times the next input. The next input is H z plus K y tilde. Substituting gives D H z plus D K y tilde.

Slide 14 | continued

Hit and miss matrices in the paper. Stay on this deck slide until the explicit click cue.

07 [Sweep across the full second row from left to right.]

Collect the terms in the state-vector order: C A, D K, D H, C B. These products come from substitution, not from a different controller or an extra attack mechanism.

08 [Move to the second and third rows of Φ_m on the right.]

In the miss matrix, the identity blocks copy the stored measurement and controller memory unchanged. This is precisely what we meant by a frozen computation state.

09 [Point to Δ in the bottom-right corner.]

Only the actuator-memory block depends on HOLD or ZERO. If Delta is the identity, the old input is copied. If Delta is zero, the next input is zero.

10 [Look at the audience.]

Imagine leaving a pump at its previous setting while its next calculation is delayed. HOLD preserves that setting. ZERO removes the model's input deviation. Which is physically preferable depends on the plant and operating point.

11 [Point to the bottom switched-model box.]

Equation four selects between the two matrices. Equations five and six define them. Equation seven, which we see next, is the optimisation problem containing these dynamics.

12 [Pause for two seconds.]

The complete state has the dimensions of x plus y plus z plus u. Each block has to map the matching component. This gives us a useful structural check before using any numerical implementation.

13 [Click once to slide 15.]

Now that we can predict the effect of any miss pattern, we need to say what makes one pattern worse than another.

Slide 15 | From objective to effectiveness ω

Swarali Nilesh Savale / Paper Eq. (7), Eq. (9), Eqs. (10)-(11); Definition 5.1

AT A GLANCE / do not read this block separately

```
J_attack = y_ℓ^T Q y_ℓ
ω = J_attack - J_nominal
Choose a binary sequence, subject to dynamics, watchdog and statistical constraints.
```

01 [Point to the maximisation at the top-left.]

The attacker chooses the sequence a zero through a ell minus one. Its objective is the terminal output cost, y at ell transpose Q y at ell. Ell is a fixed, finite horizon for this optimisation.

02 [Point to Q in that expression.]

Q is a fixed positive-semidefinite weight chosen before the search. It specifies what matters physically. On Furuta, the cost selects the pendulum angle. On the tank, it scores both regulated outputs.

03 [Point to the binary constraint and the specified initial state.]

The decisions are hits or misses, and the initial augmented state is fixed. Gurobi is not allowed to create a more damaging initial condition or directly choose y at ell.

04 [Point to the sum ending at j+a-bar.]

This sum covers a bar plus one consecutive slots and allows at most a bar misses. That guarantees at least one hit in each such group and prevents a longer consecutive run.

05 [Point to the switched-dynamics line.]

The next constraint propagates the actual state using the selected hit or miss matrix. Every candidate attack must obey these dynamics.

Slide 15 | continued

From objective to effectiveness ω. Stay on this deck slide until the explicit click cue.

06 [Point to the last two constraints.]

The remaining lines enforce statistical stealth: the permitted miss count in each valid binomial window, and the geometric chi-square acceptance condition. We will unpack both tests next.

07 [Move to J_attack and ω in the right-hand box.]

The maximised quantity is J attack. The reported effectiveness, omega, subtracts the nominal terminal cost. Because the nominal run uses the same initial state, that subtracted value is constant during this search.

08 [Look at the audience; pause after the first sentence.]

So the sequence that maximises J attack also maximises omega. But the meanings differ: J attack is the final weighted deviation, while omega is the additional final cost relative to nominal control.

09 [Point to the limitation box.]

Imagine comparing two trips only by the distance from the destination at the final instant. A dangerous detour earlier in the trip might not appear in that final score. Likewise, this objective does not score the largest transient excursion.

10 [Point back to the finite-horizon heading.]

The optimisation therefore gives a worst case for this model, initial state, horizon, cost and active detector. It is not automatically an infinite-time stability statement or a physical-safety certificate.

11 [Click once to slide 16.]

We have described the attacker’s objective. Let us now see exactly what the first two timing restrictions allow.

Slide 16 | Detection I: watchdog and binomial test

Swarali Nilesh Savale / Paper Definition 4.1; Section 4.4.1; Eqs. (8)-(9)

AT A GLANCE / do not read this block separately

```
Watchdog: no more than a-bar consecutive misses
S_w = sum of misses in a window of n_w jobs
S_w ≤ B_(1-α)(n_w, p-hat-m)
```

01 [Point to the watchdog box on the left.]

The watchdog is a local burst rule. If a bar is two, a run of three misses triggers the rule. But miss, hit, miss, hit can continue without ever producing three consecutive misses.

02 [Hold up two fingers, then point to the binomial box.]

That example explains why we need another view. The binomial test counts all misses inside the most recent window, even when hits separate them.

03 [Point to w_i and S_w.]

Each w is zero or one. Summing these bits gives S sub w, the number of misses in the window. N sub w is the window length, not the number of consecutive misses.

04 [Point to the model-assumption box, then the Binomial expression.]

Under the nominal independent-miss model, this count follows a binomial distribution. P hat sub m is estimated from normal timing data. It supplies the reference against which extra misses are judged.

05 [Point to H_0 and H_1.]

The null says the miss probability is no greater than the nominal reference. The alternative says it is greater. This is a one-sided test because the suspicious event here is an excessive miss count.

Slide 16 | continued

Detection I: watchdog and binomial test. Stay on this deck slide until the explicit click cue.

06 [Point to the acceptance inequality.]

The quantile on the right is a critical count. A sequence stays acceptable when its window count does not exceed that threshold, using the convention implemented in the paper’s code.

07 [Look at the audience and give a concrete example.]

For example, with a window of twenty, nominal miss probability one quarter and alpha one percent, the REP quantile is ten. The nominal expected count is five, but five is not the rejection threshold.

08 [Point again to the inequality.]

The threshold allows for natural statistical variation. Under this convention, ten misses pass the count constraint and eleven fail it. We check every applicable sliding window, not just disjoint groups of twenty.

09 [Point to the lower distinction box.]

Imagine an attendance monitor. One rule checks whether absences happen several days in a row. Another checks the total absences over a recent period. Neither rule is a substitute for the other.

10 [Pause for two seconds.]

Passing the test means we fail to reject the timing record under the assumed nominal model. It does not establish that the sequence is innocent, or that an attacker was absent.

11 [Click once to slide 17.]

Even with both of these restrictions, an attacker may arrange the permitted misses in an unusual pattern. That motivates the geometric test.

Slide 17 | Detection II: the geometric (χ^2) test

Swarali Nilesh Savale / Paper Section 4.4.2; Eqs. (10)-(11); REP interval-count convention

AT A GLANCE / do not read this block separately

```
For r < a-bar: p_r = (p-hat-m)^r (1-p-hat-m); final p_r = (p-hat-m)^(a-bar)
E_r = N p_r; S_chi-square = sum_r (m_r-E_r)^2 / E_r
```

01 [Point to the run-length definition at the upper left.]

Instead of asking only how many misses occurred, we now ask how they were arranged. R counts the misses between one completed hit and the next hit. Adjacent hits belong to category zero.

02 [Point to the geometric probability for r below a-bar.]

For r misses followed by a hit, independence gives p to the power r, multiplied by one minus p. A longer burst is less common under the nominal model.

03 [Point to the final category.]

The last category pools the geometric tail at a bar. That is why its probability is p to the power a bar, rather than that value multiplied by one minus p. We follow the paper and code's category convention.

04 [Point to N and E_r.]

M sub r is the observed count. N is the number of counted intervals, and E sub r is N times the category probability. We compare observed counts with these expected counts.

05 [Point to the statistic on the right.]

Each term measures squared disagreement, divided by its expected count. A large total means the observed pattern is far from the nominal geometric pattern.

06 [Point to the denominator E_r.]

This division also explains why rare bursts matter. A large discrepancy in a category expected to occur only rarely can contribute strongly to the statistic.

07 [Point to the acceptance limit.]

The statistic must stay below the chi-square critical value. The paper uses a bar degrees of freedom because there are a bar plus one categories and the total count is fixed, treating the nominal probability as fixed for this check.

Slide 17 | continued

Detection II: the geometric (χ^2) test. Stay on this deck slide until the explicit click cue.

08 [Look at the audience; slow down for the example.]

Here is a small example, using REP boundary hits. Surround the interior sequence one one zero one zero with a hit at each end. The completed intervals contain two misses, one miss and zero misses. Therefore N is three.

09 [Point back to m_r and E_r.]

With p equal to one quarter and a bar equal to three, the observed counts are one, one, one, zero. The expected probabilities are point seven five, point one eight seven five, point zero four six eight seven five, and point zero one five six two five.

10 [Point to S_chi-square.]

Substituting gives S approximately six point three three. With alpha one percent and a bar three, the critical value is about eleven point three four five. This example passes.

11 [Briefly refer back to the glossary without changing slides.]

If the code instead displays nineteen against about thirty-four, it has multiplied both sides by N, which is three. The decision is the same. A different a bar also changes the degrees of freedom and the threshold.

12 [Point to the alert box at the lower right.]

The watchdog limits consecutive misses. The binomial test limits the recent total. The geometric test checks the distribution of intervals. Together, they restrict different ways an attack can hide in normal timing behaviour.

13 [Look at the audience.]

At this point, statistical acceptance is a property of the chosen tests. The chi-square calibration is approximate, especially with small expected counts. That is one reason a defender study also needs empirical false-alarm measurements.

14 [Say the hand-off, then click once to slide 18 and step aside.]

We now have the complete chain: a timing decision selects a state update, the state update determines physical impact, and the detectors constrain the allowed sequence. I will hand over to Madhavi Singh to show what this produces on the paper’s two experimental plants.

Srinjoy Chakraborty

Policy replay, defender study and closing

Deck slides 34-42

Start after Abhishek’s reproduction result. Your main message is that a controlled replay, an optimised worst case and a validated defender ranking require different evidence.

Your sequence: divider; published HOLD replays; published ZERO and extra cases; tank detector comparison; tank policy comparison; defender formulation; missing-horizon result; shared Conclusion; References.

Full delivery: about 23-26 minutes. The following pages are the complete spoken script.

Slide 34 | Visualization Cases and Defender Extension

Srinjoy Chakraborty / Opening hand-off; paper Section 5 and project extension

01 [Face the audience after Abhishek finishes. Keep slide 34 visible.]

Thank you, Abhishek. I am Srinjoy Chakraborty. Abhishek showed our independent reproduction. I will now explain what happens when we replay selected attacks under different actuator policies, and how that leads to a defender-side study.

02 [Point to “Results replay and validation”, then “defender configuration”.]

There are two levels here. A replay asks what one specified schedule does. A worst-case study asks what the strongest feasible schedule can do against a chosen defence. I will keep those questions separate.

03 [Look at the audience.]

Imagine two emergency driving policies: keep the last steering correction, or return the correction to zero. Testing both against one road event is useful. But it does not prove which policy is better against every possible event.

04 [Pause for two seconds.]

Our completed work includes the numerical replays and a partial binomial sweep. The full detector comparison and session false-alarm calibration remain planned. I will identify the missing evidence rather than present the partial study as a completed ranking.

05 [Click once to slide 35.]

Let us start with the three Furuta attacks published in the paper’s Table one.

Slide 35 | Furuta: the three published HOLD attacks

Srinjoy Chakraborty / Paper Table 1 / Fig. 4; our same-schedule REP replay

AT A GLANCE / do not read this block separately

```
ℓ = 25; T_s = 5 ms; a-bar = 4; p-hat-m = .25; α = .01
ω = J_attack - J_nominal; χ²_(.99)(4) ≈ 13.277
```

01 [Point to the settings line at the top.]

These are twenty-five-period attacks with a five-millisecond sample time. The replay spans one hundred and twenty-five milliseconds after the initial forced hit. Each policy starts from the same initial state and receives the same miss schedule.

02 [Point to the plot axes before pointing to a curve.]

The horizontal axis is time in seconds. The vertical axis is pendulum angle in degrees. These traces describe the model's motion; the table below uses a terminal cost in radians squared.

03 [Point to the three HOLD legend entries, then follow each curve near the right edge.]

The three HOLD trajectories correspond to binomial windows of ten, twelve and fourteen jobs. Their largest absolute angles here are about twenty-one point seven five, sixteen point zero three and seventeen degrees.

04 [Point to the dashed ZERO curve and its legend.]

The dashed ZERO trace shown is the window-ten schedule replayed under ZERO. The other ZERO outcomes appear in the table. This is a policy change for a fixed attack, not a newly optimised ZERO attack.

05 [Point to the first table row, Window 10 and Misses 15.]

For the window-ten case, the full horizon contains fifteen misses. That does not mean fifteen misses occur inside one ten-job window. Its critical window allowance is six; the schedule distributes the misses accordingly.

06 [Point to ω_HOLD = 0.144120 in the same row.]

Its additional terminal angle cost under HOLD is about zero point one four four one two radians squared. The other HOLD costs are about zero point zero seven eight two two two and zero point zero eight seven nine eight six.

Slide 35 | continued

Furuta: the three published HOLD attacks. Stay on this deck slide until the explicit click cue.

07 [Point to the negative ZERO values. Pause before explaining them.]

The ZERO differences are slightly negative, around minus seven times ten to the minus six. Negative omega means the terminal cost is smaller than the nominal run’s terminal cost at that sample. It does not mean a negative squared angle.

08 [Look at the audience.]

A fixed sequence designed to hurt HOLD may accidentally leave a smaller terminal error under ZERO. That does not make it a protective attack, and it does not tell us the worst attack against ZERO.

09 [Point to the geometric-test results beside the screenshot.]

All three schedules pass the watchdog and their binomial constraints. But their geometric chi-square statistics are about one hundred and eighty-seven, eighty-six and ninety-five. All exceed thirteen point two seven seven.

10 [Pause, then point to the same threshold again.]

Therefore these are stealthy against the active binomial test, but they would fail the combined detector. There is no contradiction: adding another test changes the feasible attack set.

11 [Point briefly to the dashboard image. Do not attempt to operate it.]

This is an offline dashboard screenshot used to illustrate the angle view. Our evidence here is numerical replay. We are not claiming a new hardware experiment or inferring physical toppling from a particular angle in this local model.

12 [Click once to slide 36.]

So policy matters, and the active detector matters. Next I will compare schedules selected for the ZERO policy and different horizons.

Slide 36 | Furuta: horizon and detector validation

Srinjoy Chakraborty / Paper Section 5.4.1 / Fig. 2; extra project replay

AT A GLANCE / do not read this block separately

```
Published ZERO schedules: ℓ=5: 11010; ℓ=15: 101011011101000
ℓ=25: 1100000000010110111011010
Standard S_chi-square limit = 13.277 for a-bar=4, α=.01
```

01 [Point to the opening two lines.]

The three published ZERO cases use the combined detector and a window of twenty. The additional twenty-three-period case uses a binomial window of ten. They solve different attack problems, so we should not rank them solely by horizon.

02 [Point to the plot axes and the three horizon labels.]

The plot shows ZERO angle in degrees over physical time. The shorter trajectory ends at its own horizon. Five, fifteen and twenty-five periods correspond to twenty-five, seventy-five and one hundred and twenty-five milliseconds.

03 [Point to the ZERO column for ℓ=5, 15 and 25.]

Their additional terminal costs are about two point seven four, eight point four six, and two point eight eight times ten to the minus five radians squared. The fifteen-period case is larger than the twenty-five-period case.

04 [Look at the audience.]

Think of pushing a swing. The effect depends on when the pushes occur and when you measure the result, not only on how long you have been pushing. Here the objective scores one terminal sample.

05 [Point to the HOLD column in those same three rows.]

These HOLD values reuse the ZERO-selected schedules. They answer a counterfactual replay question. They are not independently optimised HOLD worst cases.

Slide 36 | continued

Furuta: horizon and detector validation. Stay on this deck slide until the explicit click cue.

06 [Point to the line listing 6.33, 12.83 and 12.11.]

All three published schedules pass the geometric part of the combined detector. Their statistics are below thirteen point two seven seven. They also satisfy their watchdog and binomial constraints.

07 [Point to 6.33; refer to Swarali by name.]

Swarali’s small example also gave six point three three. Her a bar of three used a threshold of eleven point three four five. Here a bar is four, so the degrees of freedom and threshold differ. The statistic is not being compared inconsistently.

08 [Point to the Extra, ℓ=23 row.]

The extra case produces about zero point zero four two four additional cost under HOLD and a small negative difference under ZERO. Its geometric statistic is one hundred and twenty-six point one one, so it fails the combined detector.

09 [Point to the feasibility caveat below the table.]

That extra schedule is feasible for its binomial setting, but we do not have a proof that it is optimal. A feasible damaging example and a certified worst case are different strengths of evidence.

10 [Point to the bottom caption, then look at the audience.]

The lesson is that horizon, policy, initial state and detector jointly determine the attack problem. A longer horizon alone does not imply a larger terminal cost.

11 [Click once to slide 37.]

To check whether this reasoning is specific to the unstable pendulum, let us turn to the stable quadruple tank.

Slide 37 | Quadruple tank: detector comparison and replay

Srinjoy Chakraborty / Paper Figs. 2-3; exact REP tank matrices; independent finite enumeration

AT A GLANCE / do not read this block separately

```
ω_bin = .003161519; ω_combined = .002073517
Reduction = (ω_bin - ω_combined) / ω_bin ≈ 34.4%
```

01 [Point to the settings line.]

The tank uses the paper’s HOLD model with a sample time of zero point one seconds. Here we compare two ten-period searches. Because there are ten free binary decisions, all one thousand and twenty-four schedules can be enumerated.

02 [Point to the horizontal and vertical axes.]

The horizontal axis is seconds. The vertical axis is the lower-tank output y two. These values are deviations from equilibrium, not absolute water heights and not evidence of physical overflow.

03 [Point to the nominal dashed trace, then the two ten-period traces.]

The nominal trajectory regulates toward zero. With binomial-constrained misses, the output decays more slowly. Adding the geometric constraint limits the available attacks, so the combined-search trajectory lies closer to nominal for this optimum.

04 [Point to the two costs in the table.]

The binomial-only maximum is about zero point zero zero three one six two. The combined maximum is about zero point zero zero two zero seven four. That is a reduction of about thirty-four point four percent in this finite-horizon configuration.

05 [Point back to the y_2 axis, then to the cost column.]

There is a small but important distinction: the plot shows one output component. The reported terminal cost weights both lower-tank outputs. We cannot recover the whole cost just by reading the vertical gap of this one curve.

Slide 37 | continued

Quadruple tank: detector comparison and replay. Stay on this deck slide until the explicit click cue.

06 [Point to the χ^2 column.]

The binomial-only optimum has chi-square about three hundred and thirty-eight point seven eight. It passes its active binomial check but fails the geometric check. The combined optimum has chi-square about ten point seven one and passes the thirteen point two seven seven limit.

07 [Point to the Extra, ℓ=23 row.]

The extra twenty-three-period schedule gives about zero point zero zero one five four nine, with chi-square fifteen. It passes its binomial setting and fails the geometric test. It is not a proven optimum.

08 [Look at the audience.]

This is what adding a constraint should do to a maximisation problem: at fixed settings, the best feasible attack cannot become larger when the feasible set shrinks. Here we also know the numerical optima because the ten-bit search is exhaustive.

09 [Pause for two seconds.]

However, equal per-test alpha does not establish equal false-alarm probability over a whole session. The thirty-four percent comparison is a constrained-cost result, not yet an equal-false-alarm operational ranking.

10 [Click once to slide 38.]

The remaining policy question is whether ZERO, which looked attractive on Furuta, also helps on this tank.

Slide 38 | Quadruple tank: policy comparison under a replayed schedule

Srinjoy Chakraborty / Project ZERO extension; same schedules and initial states

AT A GLANCE / do not read this block separately

```
y_k = .5 [x_1,k, x_2,k]^T
ω_output = .25 ω_REP
A fixed-schedule comparison does not replace separate worst-case optimisation.
```

01 [Point to the first two sentences.]

For every bar pair, the initial condition and miss schedule are identical. HOLD is the paper’s tank model. ZERO on the tank is our project extension, implemented by changing the miss-mode actuator-memory block.

02 [Point to the legend, then the first pair of bars.]

The first pair uses the ten-period binomial schedule. Its HOLD cost is about zero point zero zero three one six two, while ZERO gives about zero point zero zero nine two three one. The ZERO value is about two point nine two times as large.

03 [Point to the second and third pairs.]

The corresponding ratios for the combined ten-period schedule and the extra twenty-three-period schedule are about two point one two and two point four eight. All three fixed schedules give a larger terminal cost under this ZERO extension.

04 [Look at the audience.]

Imagine a tank operating around a normal pump setting. Removing the corrective input deviation can remove a useful correction. For a pendulum, continuing an old motor correction can instead be harmful. The plant dynamics decide the consequence.

Slide 38 | continued

Quadruple tank: policy comparison under a replayed schedule. Stay on this deck slide until the explicit click cue.

05 [Point to the paragraph beginning “Tank ZERO”.]

Zero here means zero input deviation in the linearised operating-point model. It does not mean the physical pumps have been switched off. We should not translate a zero in the model into that hardware claim.

06 [Point to the output-scaling equation.]

The paper defines the two tank outputs as one half of the corresponding physical state components. Squaring a factor of one half gives one quarter. Therefore the output-cost difference is the raw REP state-weighted difference divided by four.

07 [Point to the interpretation paragraph.]

These replays show that ZERO is not universally beneficial across plants and schedules. They give a concrete counterexample to transferring the Furuta observation as a universal rule.

08 [Point to the final block about ranking policies.]

To rank policies by worst-case cost, we must let the attacker optimise separately against each policy. These schedules were selected or found for HOLD. Keeping them fixed gives a controlled replay comparison, not the complete min-max comparison.

09 [Pause, then click once to slide 39.]

That leads naturally to the defender’s problem: rather than accepting one fixed configuration, can we choose a configuration whose strongest accepted attack is less damaging?

Slide 39 | Defender study: formulation and completion status

Srinjoy Chakraborty / Project formulation; partial binomial results; planned extensions

AT A GLANCE / do not read this block separately

```
d*(p-hat-m) = arg min_d max_(ℓ in L) ω*(d, p-hat-m, ℓ)
subject to FA(d, p-hat-m) ≤ ϕ
```

01 [Point to p-hat-m in the opening sentence.]

We solve this problem separately for each measured nominal miss scenario. P hat sub m describes the system’s normal timing. It is not a protection setting that we can simply reduce on paper.

02 [Point to ω-star in the inner expression.]

Read the formula from the inside. Omega star already means the strongest feasible attack for a fixed defence and a fixed horizon. The attacker has optimised the miss sequence.

03 [Point to the maximum over ℓ.]

The next maximum compares the tested horizons and keeps the largest impact. A defender should not look safe merely because we happened to test an easy attack length.

04 [Point to arg min over d.]

The outer minimisation chooses a defence configuration: actuator policy, detector, relevant window length and significance level. We want the smallest worst attack within the declared experiment.

05 [Point to the false-alarm constraint.]

But a detector that constantly alarms is not a useful defence. We therefore require the probability of an alarm during a nominal session to stay below a specified limit phi.

06 [Look at the audience.]

Imagine a smoke alarm. Making it extremely sensitive may detect more problems, but it becomes unusable if normal cooking repeatedly sets it off. Physical protection and operational false alarms must be considered together.

Slide 39 | continued

Defender study: formulation and completion status. Stay on this deck slide until the explicit click cue.

07 [Point down the Status column.]

The status column is essential. Policy replays are done. The binomial sweep is partial. The geometric and combined-detector sweeps are planned. Horizon twenty-five is missing, and empirical session false-alarm estimation is also planned.

08 [Point to the false-alarm block.]

The proposed calibration simulates nominal Bernoulli timing sessions of a declared length. We count the sessions with at least one alarm, report uncertainty, and specify whether watchdog shutdowns count. This experiment has not yet been completed.

09 [Point to α in the table.]

Alpha is the level of an individual test. It is not automatically the probability of at least one alarm over many repeated, overlapping checks. That is why the session-level constraint is written separately.

10 [Point to the grid-count block.]

For the current binomial design, two policies, six windows and three alpha values give thirty-six defender choices per scenario. Three scenarios and five horizons give five hundred and forty calls; twenty-five horizons give two thousand seven hundred.

11 [Point to the planned counts in the same block.]

With all three detectors, the planned counts are one thousand one hundred and seventy quick calls or five thousand eight hundred and fifty full calls, provided the window-independent geometric-only calculations are not repeated for every window.

12 [Look at the audience.]

The formulation is a study design, not a completed certified ranking. Any recommendation must also check horizon coverage, solver status, replay consistency and the false-alarm requirement.

13 [Click once to slide 40.]

The last result shows exactly why incomplete coverage can change the conclusion.

Slide 40 | Defender validation: horizon coverage changes the conclusion

Srinjoy Chakraborty / Saved partial notebook results; Gurobi error 10010; Table 1 replay check

AT A GLANCE / do not read this block separately

```
Available horizons: {5,10,15,20}; horizon 25 missing
.144120 / .021973 ≈ 6.56
Full-horizon worst case is at least the cost of any feasible known attack.
```

01 [Point to the first two lines before discussing the plot.]

This plot shows HOLD with a bar four, alpha one percent and the binomial detector. Each point is a maximum over the available horizons five, ten, fifteen and twenty. The horizon-twenty-five runs are missing.

02 [Point to the x-axis, y-axis and three legend entries.]

The horizontal axis is the binomial window length. The vertical axis is the maximum additional terminal cost over those available horizons. Each curve is a different measured nominal miss scenario.

03 [Trace the p-hat-m=.25 curve across the six windows.]

The red point-two-five curve decreases overall across part of the range, but rises again at window twenty. The relationship is not monotone. Window length changes both the allowed density and the possible placement of miss bursts.

04 [Point to the comparison table on the right.]

Now fix the window at ten and keep the same policy, nominal probability, alpha and detector. The saved maximum over available horizons is about zero point zero two one nine seven three.

05 [Point to the Table 1 replay value.]

But a known feasible twenty-five-period attack from Table one produces about zero point one four four one two zero. That is roughly six point five six times the displayed partial maximum.

Slide 40 | continued

Defender validation: horizon coverage changes the conclusion. Stay on this deck slide until the explicit click cue.

06 [Look at the audience; pause after the next sentence.]

This ratio compares additional squared-angle costs, not angles in degrees. It demonstrates that the truncated maximum substantially understates a damaging attack already known at the missing horizon.

07 [Point to the remaining-work paragraph.]

The saved log explains the failure: Gurobi error one zero zero one zero, model too large for the size-limited licence. This is not a timeout and does not show that the attack is infeasible.

08 [Keep the pointer on that paragraph.]

A correctly configured unrestricted academic licence removes this size restriction. It does not automatically remove the computational difficulty or guarantee every solve finishes with a proof of optimality.

09 [Point to “432/540” near the bottom.]

Across the complete quick-grid attempt, four hundred and thirty-two rows were saved and one hundred and eight horizon-twenty-five calls failed. The plot is one HOLD and alpha-one-percent slice of that campaign, not all four hundred and thirty-two runs.

10 [Look at the audience.]

A feasible known attack is a lower bound on the true maximum damage. Therefore we already know this configuration can reach at least zero point one four four one two. We cannot use the smaller truncated result as a safety guarantee.

11 [Pause for two seconds, then click once to slide 41.]

The supported result is that configuration choices change observed attack impact. The remaining work is to complete the horizons, add the other detectors and measure session false alarms before ranking the defences. Let me close with the overall findings.

Slide 41 | Conclusion

Srinjoy Chakraborty / Shared group closing; paper findings and project scope

01 [Point to the left block.]

The paper’s central message is that deadline misses are a security surface even without forged measurements or commands. An attacker can exploit the difference between a physically correct command and a command delivered at the right time.

02 [Point to the statistical-stealth bullet.]

Stealth means that the active statistical tests fail to reject the sequence. It does not mean that no possible detector could detect it.

03 [Point to the plant/policy bullet and then the finite-horizon bullet.]

The impact depends on the plant, schedule and actuation policy. The paper formulates a finite-horizon worst-case problem and demonstrates severe Furuta disruption. Those conclusions remain tied to the stated model and detector constraints.

04 [Move to the right block.]

Our group independently reproduced the reduced Furuta curve, replayed the published Table one schedules, and showed that all three fail the geometric test even though they pass the binomial constraints they were designed against.

05 [Point to the policy-replay and defender-study bullets.]

We also showed why a fixed-schedule policy comparison cannot establish a universal ZERO rule, and formulated the defender-side min-max study with partial binomial results.

06 [Point to the limitations block.]

Our limitations are explicit: a terminal-cost objective, independent nominal misses in the reference model, no project hardware experiment, missing horizon-twenty-five solves and unfinished session false-alarm calibration.

07 [Face the audience; pause before the last sentence.]

The practical conclusion is that timing monitoring, actuator fallback and controller dynamics must be evaluated together. A detector setting should be recommended using complete evidence at an acceptable false-alarm rate.

08 [Click once to slide 42.]

These are the main references and the official repeatability resources.

Slide 42 | References

Srinjoy Chakraborty / Paper DOI, official REP v1.2 and taxonomy reference

01 [Point to reference 1, then reference 2.]

The first reference is the ICCPS twenty-twenty-five paper by Nauta, Sandberg and Maggio. The second is their official repeatability package, which supplies the system matrices and optimisation implementations.

02 [Point to reference 3.]

The Teixeira reference is the source of the resource-based attack taxonomy shown earlier in the introduction.

03 [Point briefly to reference 4; do not click placeholder links.]

Our project-link entries are still explicitly marked as editable placeholders in the glossary. The numerical evidence discussed today is identified by the paper, code and replay descriptions rather than those dummy addresses.

04 [Move the pointer away, face the audience and stop advancing slides.]

Thank you for listening. We are happy to take questions.

Ready answers if the audience asks

These answers are separate from the scripted presentation. They add technical precision without making the live explanation unnecessarily dense. Each answer is written as something you can say aloud.

Swarali answers questions about the model, notation and statistical constraints. Srinjoy answers questions about the replay evidence, policy comparison and defender-study scope.

Swarali Nilesh Savale | question bank

01. Does the attacker need the model or not?

Execution and planning require different capabilities. A precomputed schedule can be executed without reading sensor values. Computing the worst-case schedule in this paper uses the plant/controller model, initial augmented state and detector settings. We should not describe the optimiser as model-free.

02. Can any low-priority task starve a controller?

No. The attacker needs an actual mechanism for delaying the job, such as scheduling priority, resource blocking or an implementation-specific interference path. The paper discusses starvation mechanisms, but merely sharing a processor is not a sufficient capability assumption.

03. Is the delayed job dropped?

No, not in this paper's model. The delayed computation continues and completes using the stored measurement. Controller memory and the stored measurement remain frozen during misses. A system that drops and restarts late jobs requires different hit/miss dynamics.

04. Why does the hit matrix use y-tilde before refreshing the buffer?

The augmented state is recorded at a specific point in the execution cycle. The completing computation uses the measurement associated with that job. Its command and controller state update from that stored value; the next buffer entry becomes the next output. The CA, DK, DH and CB row follows by substituting x_next and u_next into y_next=C x_next+D u_next.

05. What dimensions should the blocks have?

Let x have n_x entries, y have n_y, z have n_z and u have n_u. Then psi has n_x+n_y+n_z+n_u entries. A is n_x by n_x, B is n_x by n_u, C is n_y by n_x, D is n_y by n_u, F is n_z by n_z, G is n_z by n_y, H is n_u by n_z and K is n_u by n_y. For example, DK must map stored y to next y, so it is n_y by n_y.

06. Why can two schedules with the same number of misses cause different damage?

The final state depends on the ordered product of hit and miss matrices acting on the initial state. Matrix products generally do not commute. A miss placed before or after a useful correction can therefore have different consequences, even when total count and longest burst are identical.

07. Does a stable hit matrix imply safety under misses?

No. Nominal closed-loop stability is not a guarantee for every switched hit/miss sequence. We need to analyse the allowed products, or an appropriate switched-system stability condition. The finite-horizon optimisation evaluates the chosen terminal cost; it does not prove arbitrary-switching or infinite-horizon stability.

08. Why is the binomial threshold ten rather than the expected count five?

Five is the mean of a Binomial(20,.25) count. A rejection threshold must also allow statistical variation. The REP convention uses the .99 binomial quantile, which is ten. Its inequality allows ten and rejects counts above ten. Discreteness makes exact tail-test boundary conventions important; I am stating the implemented convention.

09. What does N count, and what if there are no intervals?

With REP's forced boundary hits, N is the number of completed hit-to-hit intervals counted by the pattern constraints. It is one fewer than the number of hit markers, not the total number of periods. A textbook chi-square statistic is undefined when N is zero. The project audit explicitly handles this degenerate case by setting its informational statistic to zero; that convention is not evidence of statistical power.

10. Why is the final geometric probability p to the power a-bar?

The final category pools the geometric tail at the maximum recorded category. The probabilities below it sum to one minus p to the power a-bar, leaving p to the power a-bar for the last category. This follows the paper/REP convention. Deployment still needs calibration with the actual watchdog and boundary rules.

11. Why are the thresholds 11.345 and 13.277 both used?

At alpha=.01, the paper uses chi-square degrees of freedom equal to a-bar. A-bar three gives about 11.345, while a-bar four gives about 13.277. Separately, REP can compare T=N*S to N times the critical value. We must keep both the degrees of freedom and the scaled-versus-unscaled convention consistent.

12. Is the chi-square test exact for these short sequences?

No. The paper uses an approximate goodness-of-fit test. With tiny expected category counts, the usual chi-square approximation can be poorly calibrated. The numerical replay checks the specified algebraic constraints, rather than proving an exact false-alarm rate. Nominal session simulation, or another calibrated test, is needed before an operational comparison.

13. What if the nominal miss probability is estimated from the same sample?

The paper/REP comparison treats the nominal miss probability as fixed when selecting the stated critical value. Estimating parameters from the tested counts can change the reference distribution and degrees of freedom. We should not silently reuse the same calibration. A practical detector should separate calibration data or validate the complete estimation-and-testing procedure.

14. Does passing all tests prove there was no attack?

No. It means the active tests failed to reject the sequence under their assumptions. There can be malicious sequences inside the acceptance region. This is exactly the region the optimisation searches. A different detector, or a different nominal model, can make a different decision.

Srinjoy Chakraborty | question bank

01. How can a squared-angle cost be negative?

The squared-angle cost itself is non-negative. Omega is the difference between the attack and nominal costs, and a difference can be negative. In these policy replays, the fixed schedule chosen for the other policy happens to leave a smaller terminal cost. That does not imply smaller error at every earlier sample or a negative physical angle cost.

02. Why not conclude ZERO is the better Furuta defence from the replay table?

The controlled replay isolates what a policy change does to one schedule. Worst-case ranking requires a new strongest attack under each policy, with comparable detectors and initial conditions. The paper's independently optimised Furuta comparison is separate evidence. Our fixed-schedule table alone cannot supply a ZERO worst-case bound.

03. Do the 21.75 degrees prove that the pendulum fell?

No. They describe the numerical trajectory of the supplied local model. We did not define and validate a hardware failure threshold or perform a new physical experiment. A large excursion can also leave the region where a linearised model accurately represents the plant. The paper's physical experiments are distinct from our replay.

04. Why are Table 1 attacks rejected by chi-square if the paper calls them stealthy?

They were optimised against the binomial detector and watchdog. The geometric test was not active in that problem. The same schedules can therefore pass the original acceptance constraints and fail a stronger combined set. Stealth is always relative to the active detector and its settings.

05. Are the extra 23-period schedules actually 25 periods?

The interior attack horizon is twenty-three. REP-style storage adds a hit at the start and another at the end, so twenty-five stored decisions can represent twenty-three free attack periods. For a genuine twenty-five-period attack, the stored sequence has twenty-seven decisions. Our replay starts after the initial forced hit and scores before the final forced hit.

06. Why does the tank graph show y2 if the cost uses two outputs?

The graph visualises one component for clarity. The objective weights both lower-tank outputs. The numerical omega values come from the full declared output cost, not from the plotted y2 component alone. A small-looking separation of y2 curves does not by itself determine the total cost difference.

07. Does 34.4 percent prove that the combined detector is universally better?

No. It is the reduction in the exhaustive ten-period optimum for the specified settings. Adding a constraint shrinks that feasible set. It does not establish operational superiority at equal session false-alarm rates, nor a universal ordering across plants and horizons. That calibration is planned.

08. Does tank ZERO mean turning off the pumps?

No. The linear model uses input deviations around an operating point. ZERO sets that deviation command to zero on a miss, so the physical nominal operating input is not necessarily zero. The project extension changes the actuator-memory block in the miss matrix; it is not a direct claim about a hardware pump-off command.

09. Why divide the tank REP cost by four?

The plotted paper output is one half of the first two physical state components. Its quadratic output cost is therefore one quarter of the corresponding raw state-weighted REP cost. Applying the same scaling to attack and nominal costs also scales their difference by one quarter.

10. Can we select a smaller p-hat-m to make the detector better?

We cannot honestly choose the measured nominal timing rate as a protection knob. It must reflect calibration data and its uncertainty. The defender can choose policy, detector settings and possibly scheduling improvements that actually change timing; declaring a different probability without changing the system can miscalibrate false alarms.

11. How would you estimate session false alarms?

Choose a session length H and define exactly what counts as an alarm. Simulate many independent nominal sessions with the stated timing model, running the full monitor on every applicable window and interval count. Estimate the fraction of sessions that alarm and report a confidence interval. Real deployment also needs to test whether nominal misses are independent and stationary.

12. Why is alpha not the session false-alarm rate?

Even in the simplified case of M independent checks with false-alarm probability alpha, the probability of at least one alarm is 1-(1-alpha)^M. For 100 checks at one percent, that is about 63.4 percent. Our overlapping checks are dependent, so that calculation is only an illustration, not their calibration. It explains why session length and the full monitoring procedure matter.

13. Why were horizon-25 runs missing?

The saved notebook reports Gurobi error 10010: model too large for the size-limited licence. A correctly configured unrestricted licence removes that size limit. We should then rerun, inspect solver status and gaps, and replay any reported solution. The error is neither an infeasibility proof nor evidence of a computational timeout.

14. Can the incomplete grid identify the best configuration?

Not reliably for the full declared problem. Missing horizons can hide a much larger attack and distort a configuration ranking. The known feasible Table 1 attack already raises the demonstrated cost at the matching setting to at least .144120. Complete horizon coverage and false-alarm calibration are needed before a final recommendation.

15. Does a feasible replay give an upper or a lower bound on worst damage?

It gives a lower bound: the maximum over feasible attacks must be at least as large as the cost of this one feasible attack. A maximisation solver’s validated upper bound can bound the worst damage from above. A feasible sequence without a small optimality gap is not an upper-bound safety certificate.

16. What is the contribution if the full defender sweep is unfinished?

The completed contribution is independent reproduction and transparent policy/detector replay, including checks that expose the limits of a partial sweep. The defender min-max and session false-alarm constraint define the next study. This is a project extension of the paper’s fixed-defence analysis, not a claim of a globally novel, completed or deployment-certified defence.

One-page recovery sheet

If you lose your place, use the matching line below, point to the indicated object and then resume the next numbered step on that slide.

Swarali: the loop

“The plant continues evolving under the input already applied, while the next command is still being computed.” Point to the plant equation.

Swarali: the delayed job

“The physical state moves, but controller memory and the stored measurement remain frozen during the miss.” Point to the miss matrix.

Swarali: the objective

“The attacker chooses a feasible miss sequence to maximise a terminal output cost from a fixed initial state.” Point to the maximisation.

Swarali: the detectors

“The watchdog limits a burst, the binomial test limits a recent total, and the geometric test checks the distribution of intervals.” Point across the respective constraints.

Srinjoy: a replay

“This is the same schedule and the same initial state under two policies; it is not a fresh optimum for both policies.” Point to the table caption.

Srinjoy: detector acceptance

“These attacks pass the tests that were active when they were selected; a geometric check can still reject them.” Point to the statistic and threshold.

Srinjoy: defender choice

“For each measured nominal scenario, choose a defence that lowers the worst tested attack while meeting a session false-alarm limit.” Point to the outer minimisation.

Srinjoy: missing results

“A feasible horizon-twenty-five attack already exceeds the maximum over the available shorter horizons.” Point to the two-row comparison table.

Sources and evidence boundaries

Paper

T. Nauta, H. Sandberg & M. Maggio. Stealthy Computational Delay Attacks on Control Systems. ICCPS 2025. Sections 3-5, Eqs. (1)-(11), Definition 5.1, Figs. 2-4, Table 1 and Appendix A.

https://doi.org/10.1145/3716550.3722013

Official code

REP v1.2: define_system in main_short.py; optimise_bin.py, optimise_chi2.py and optimise_bin_chi2.py. The boundary hits, scored state, window constraints and scaled statistic follow these implementations.

https://github.com/Talitha-Nauta/REP/tree/v1.2

Archived code

Official repeatability archive, used to identify the authors' model and implementation rather than the dummy project links in the deck.

https://doi.org/10.5281/zenodo.14792914

Statistical calibration

SciPy documentation for Pearson chi-square: expected-category counts and estimated parameters affect the reference approximation. Used only for the supplementary calibration explanation.

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.chisquare.html

Project evidence

The current main.tex, case_results.csv and defender_available_summary.csv, together with the saved project notebook outputs, supply the project replay numbers and partial-grid status. These are project results, not new hardware data.

What the script adds

The balancing-stick, thermostat, pump, attendance, swing and smoke-alarm examples are teaching analogies. The matrix-row expansion, dimension check and repeated-testing example are explanatory derivations. They are not additional paper experiments or new project results.

What the script does not claim

The project replays do not establish a new hardware fall, universal policy superiority, an exact small-sample false-alarm rate, a completed detector sweep or a full-grid safety certificate. Dummy project links are not treated as evidence.